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To, The Product Science Hiring Team Oura

Subject: Application for Junior Data Scientist – Product Science

Dear Oura Hiring Team,

I am writing to express my strong interest in the Junior Data Scientist position within the Product Science team at Oura. Your mission to empower individuals through actionable health insights resonates deeply with me. My academic work and industry-facing projects have been intentionally focused on the exact foundation Oura is built upon: the intersection of advanced signal processing, machine learning, and biological time-series data.

My experience is anchored by two major, domain-specific initiatives that align perfectly with advancing your physiological feature pipeline:

- **Biotech Wearable Sensor Analytics (CLOCK Lab):** In my research at the Computational Biology lab at IITGN, I worked extensively with continuous, high-frequency smart ring time-series data provided by Ultrahuman. I designed Transformer-based architectures for multi-modal sensor data fusion to predict core physiological features, specifically Heart Rate (HR) and Heart Rate Variability (HRV), benchmarking performance against traditional Cosinor Analysis. I also formulated statistical metrics using linear regression and cross-correlations to capture and track social jet lag (SJL), compiling my findings into technical reports for internal reviews. This research is currently culminating in a formal paper publication. The strong impact of this work resulted in an offer to temporarily continue development in the wear-tech space; however, this pivotal experience solidified my desire to commit my career fully to consumer health analytics, leading me directly to Oura, the global gold standard of the industry.
- **High-Performance Signal Processing (MUSE Lab & Modsonic Instruments):** Working at the frontier of raw hardware-software integration, I engineered a real-time, FPGA-based portable accelerator pipeline for ultrasonic testing signal processing. Navigating strict hardware resource constraints required implementing dynamic programming architectures and streaming protocols to maintain a competitive 30 fps throughput. This experience solidified my ability to clean, transform, and extract meaningful metrics from highly volatile, noisy signal feeds under strict computational limits, a core constraint when managing wearable device pipelines.

In addition , my internship experience at Texas Instruments proved my ability to build scalable data infrastructure, where I optimized automated analysis pipelines across complex SQL databases to successfully process over 1 billion data points. Encountering a severe lack of metadata for a RAG pipeline, I devised a novel solution rooted in probability theory and statistical estimation. I chose to formalize and publish these findings as a research pre-print on arXiv, titled *"Sampling-Based Estimation of Jaccard Containment and Similarity."* This iterative process of translating raw, foundational engineering roadblocks into rigorous mathematical solutions is what truly drives me, prompting my intentional decision to transition from data engineering infrastructure toward core Data Science.

Oura represents the absolute pinnacle of translating multi-sensor wearable data into deep, meaningful human insights. I am incredibly eager to bring my domain experience in biological time-series data, mathematical modeling, and rigorous signal optimization to your team to help support and scale your next generation of physiological features.

Thank you for your time and consideration. I welcome the opportunity to discuss how my biotech and signal analytics background can add immediate value to Oura.

Sincerely,
Pranav Joshi